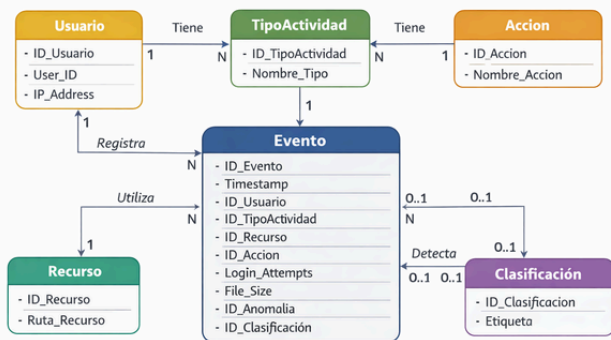


Análisis de Ciberseguridad

Modelo Entidad-Relación

Modelo Entidad-Relación Ciberseguridad



Base de datos estructurada

```
CREATE DATABASE ciberseguridad-db;  
USE ciberseguridad-db;
```

```
-- Tablas principales
```

```
CREATE TABLE Accion (  
ID_Accion INT PRIMARY KEY,  
Nombre_Accion VARCHAR(50)  
);
```

```
CREATE TABLE Anomalia (  
ID_Anomalia INT PRIMARY KEY,  
Descripcion VARCHAR(100)  
);
```

```
CREATE TABLE Clasificacion (  
ID_Clasificacion INT PRIMARY KEY,  
Etiqueta VARCHAR(50)  
);
```

```
CREATE TABLE TipoActividad (  
ID_TipoActividad INT PRIMARY KEY,  
Nombre_Tipo VARCHAR(50)  
);
```

```
CREATE TABLE Recurso (  
ID_Recurso INT PRIMARY KEY,  
Ruta_Recurso VARCHAR(200)  
);
```

```
CREATE TABLE Usuario (  
ID_Usuario INT PRIMARY KEY,  
User_ID INT,  
IP_Address VARCHAR(50)  
);
```

```
CREATE TABLE Evento (
    ID_Evento INT PRIMARY KEY,
    Timestamp DATETIME,
    ID_Usuario INT,
    ID_TipoActividad INT,
    ID_Recurso INT,
    ID_Accion INT,
    Login_Attempts DECIMAL(10,2),
    File_Size DECIMAL(10,2),
    ID_Anomalia INT,
    ID_Clasificacion INT,
    FOREIGN KEY (ID_Usuario) REFERENCES Usuario(ID_Usuario),
    FOREIGN KEY (ID_TipoActividad) REFERENCES TipoActividad(ID_TipoActividad),
    FOREIGN KEY (ID_Recurso) REFERENCES Recurso(ID_Recurso),
    FOREIGN KEY (ID_Accion) REFERENCES Accion(ID_Accion),
    FOREIGN KEY (ID_Anomalia) REFERENCES Anomalia(ID_Anomalia),
    FOREIGN KEY (ID_Clasificacion) REFERENCES Clasificacion(ID_Clasificacion)
);
```

```
INSERT INTO Accion (ID_Accion, Nombre_Accion) VALUES
    (1, 'Delete'), (2, 'Failed'), (3, 'Inserted'),
    (4, 'Read'), (5, 'Success'), (6, 'Write'), (7, 'USB_Insert');
```

```
INSERT INTO Anomalia (ID_Anomalia, Descripcion) VALUES
    (1, 'Brute_Force'), (2, 'DDoS_Attempt'), (3, 'Data_Exfil'),
    (4, 'USB_Access'), (5, 'Network_Traffic'), (6, 'Remote_Login');
```

```
INSERT INTO Clasificacion (ID_Clasificacion, Etiqueta) VALUES
    (1, 'Normal'), (2, 'Suspicious');
```

```
INSERT INTO TipoActividad (ID_TipoActividad, Nombre_Tipo) VALUES
    (1, 'File_Access'), (2, 'File_Deletion'), (3, 'File_Modification'),
    (4, 'Login'), (5, 'Network_Traffic'), (6, 'Remote_Login'), (7, 'USB_Insert');
```

Recurso (ID_Recurso, Ruta_Recurso) VALUES

(1, '/backup/data.csv'), (2, '/backup/document2.docx'), (3, '/backup/new_project.docx'), (4, '/backup/report1.pdf'), (5, '/backup/report2.pdf'), (6, '/backup/secrets.txt'), (7, '/confidential/data.csv'), (8, '/confidential/document2.docx'), (9, '/confidential/new_project.docx'), (10, '/confidential/report1.pdf'), (11, '/confidential/report2.pdf'), (12, '/confidential/secrets.txt'), (13, '/network/logs/data.csv'), (14, '/network/logs/document2.docx'), (15, '/network/logs/new_project.docx'), (16, '/network/logs/report1.pdf'), (17, '/network/logs/report2.pdf'), (18, '/network/logs/secrets.txt'), (19, '/project/data.csv'), (20, '/project/document2.docx'), (21, '/project/new_project.docx'), (22, '/project/report1.pdf'), (23, '/project/report2.pdf'), (24, '/project/secrets.txt'), (25, '/server/data.csv'), (26, '/server/document2.docx'), (27, '/server/new_project.docx'), (28, '/server/report1.pdf'), (29, '/server/report2.pdf'), (30, '/server/secrets.txt'), (31, '/shared/documents/data.csv'), (32, '/shared/documents/document2.docx'), (33, '/shared/documents/new_project.docx'), (34, '/shared/documents/report1.pdf'), (35, '/shared/documents/report2.pdf'), (36, '/shared/documents/secrets.txt');

Usuario (ID_Usuario, User_ID, IP_Address) VALUES

(1, 1000, '192.168.252.36'), (2, 1001, '172.19.88.206'), (3, 1002, '172.23.61.168'), (4, 1003, '172.28.232.9'), (5, 1004, '172.22.2.242'), (6, 1006, '172.31.86.28'), (7, 1007, '10.222.179.45'), (8, 1008, '172.19.6.195'), (9, 1010, '10.118.92.10'), (10, 1011, '172.16.180.101'), (11, 1012, '10.183.131.226'), (12, 1015, '172.16.249.225'), (13, 1018, '10.120.94.0'), (14, 1020, '192.168.19.110'), (15, 1022, '192.168.52.118'), (16, 1023, '192.168.153.106'), (17, 1024, '172.16.229.228'), (18, 1025, '10.137.44.132'), (19, 1029, '192.168.58.93'), (20, 1036, '172.18.70.213'), (21, 1037, '10.198.116.172'), (22, 1038, '192.168.2.250'), (23, 1042, '10.98.185.41'), (24, 1044, '192.168.26.206'), (25, 1045, '10.55.43.50'), (26, 1046, '192.168.168.139'), (27, 1047, '172.17.95.57'), (28, 1049, '172.25.162.160'), (29, 1052, '172.16.22.11'), (30, 1054, '192.168.102.22'), (31, 1060, '172.21.113.189'), (32, 1061, '172.22.42.87'), (33, 1062, '172.29.159.213'), (34, 1065, '172.24.203.183'), (35, 1066, '192.168.65.151'), (36, 1067, '192.168.185.139');